Homework Assignment 1: Slab-Geometry Boltzmann Equation

Tal Ramot

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1 Question 1 - Useful identity for the solution of the time-dependent slab-geometry Boltzmann equation

In this question we are asked to prove the following identity:

$$\psi(x,\mu,t;c) = ce^{-(1-c)v\Sigma_t t}\psi(cx,\mu,ct;1) \tag{1}$$

We will do it in two parts. First, we will write down the equation that $\tilde{\psi} = \psi(cx,\mu,ct;1)$ satisfies, and then we will use it in order to prove that the product $ce^{-(1-c)v\Sigma_t t}\tilde{\psi}$ satisfies the same equation as $\psi(x,\mu,t;c)$.

1.1 The equation for $\tilde{\psi}$

Notation: We will denote by $\Psi(X,\mu,T) \equiv \psi(X,\mu,T;1)$ the $c=1$ solution written in its own variables, and by Ψ_T and Ψ_X its partial derivatives with respect to them, so that $\tilde{\psi}(x,\mu,t) = \Psi(cx,\mu,ct)$ with $X = cx$ and $T = ct$. **This notation is used to avoid confusion when using the chain rule, as will be done immediately.** First, Ψ satisfies the $c=1$ equation in the variables X and T:

$$\frac{1}{v}\Psi_T + \mu\Psi_X + \Sigma_t\Psi = \frac{\Sigma_t}{2}\int_{-1}^{1}\Psi d\mu' \tag{2}$$

which after applying the chain rule, $\Psi_T = \frac{1}{c}\frac{\partial\tilde{\psi}}{\partial t}$ and $\Psi_X = \frac{1}{c}\frac{\partial\tilde{\psi}}{\partial x}$, and multiplying by c, becomes:

$$\frac{1}{v}\frac{\partial\tilde{\psi}}{\partial t} + \mu\frac{\partial\tilde{\psi}}{\partial x} + c\Sigma_t\tilde{\psi} = \frac{c\Sigma_t}{2}\int_{-1}^{1}\tilde{\psi}d\mu' \tag{3}$$

So $\tilde{\psi}$ obeys the same equation as $\psi(x,\mu,t;c)$ except that it is removed at the rate $c\Sigma_t$ instead of Σ_t.

1.2 Proving the identity

In this part we are going to simply subtitute the RHS of 1 into the LHS of the equation for $\psi(x,\mu,t;c)$ and show that it satisfies the same equation. After substitution and application of the chain rule we get:

$$\text{LHS} = \frac{1}{v}\left[c(c-1)v\Sigma_t e^{-(1-c)v\Sigma_t t}\tilde{\psi} + c\,e^{-(1-c)v\Sigma_t t}\frac{\partial\tilde{\psi}}{\partial t}\right] + \mu c\,e^{-(1-c)v\Sigma_t t}\frac{\partial\tilde{\psi}}{\partial x} + \Sigma_t c\,e^{-(1-c)v\Sigma_t t}\tilde{\psi} \tag{4}$$

$$= ce^{-(1-c)v\Sigma_t t}\left[\frac{1}{v}\frac{\partial\tilde{\psi}}{\partial t} + \mu\frac{\partial\tilde{\psi}}{\partial x} + \Sigma_t\tilde{\psi} + (c-1)\Sigma_t\tilde{\psi}\right] \tag{5}$$

$$= ce^{-(1-c)v\Sigma_t t}\left[\frac{1}{v}\frac{\partial\tilde{\psi}}{\partial t} + \mu\frac{\partial\tilde{\psi}}{\partial x} + c\Sigma_t\tilde{\psi}\right] \tag{6}$$

$$= ce^{-(1-c)v\Sigma_t t}\frac{c\Sigma_t}{2}\int_{-1}^{1}\tilde{\psi}d\mu' = \frac{c\Sigma_t}{2}\int_{-1}^{1}ce^{-(1-c)v\Sigma_t t}\tilde{\psi}d\mu' \tag{7}$$

where 3 was used in the last line, and this is exactly the RHS of the equation for $\psi(x,\mu,t;c)$.

2 Question 2 - Derivation of the time-dependent planar scalar flux for a plane source

The relation between a plane source and a point source as studied in class is given by:

$$\phi_{\text{pl}}(x) = 2\pi\int_{|x|}^{\infty}\phi_{\text{pt}}(r)rdr \tag{8}$$

The solution taken from J. C. J. Paasschens is the point-source scalar flux $\phi_{\text{pt}}(r,t)$ for $c=1$, and therefore the planar-source scalar flux is given by the integration given above. It is worth separating this from the other planar–spherical relation, $\nabla^2\phi = \frac{1}{r}\frac{\partial^2(r\phi)}{\partial r^2}$, which is an *operator* identity: it maps solutions of the source-free equation between the two geometries, and is what makes $\sin(Br)/r$ and $\cos(Bx)$ share a criticality condition in Assignment 1. It says nothing about how a *source* maps. What is needed here is the source superposition above — an infinite plane source is a continuum of point sources, so the planar Green's function is the integral of the point one over the plane. "Multiply by r" happens to give the right answer for a pure exponential, which is why the two are easy to conflate, but it fails on the Paasschens shapes.

$$\phi_{\text{pl}}(x,t) = 2\pi\int_{|x|}^{\infty}\frac{e^{-v\Sigma_t t}}{4\pi r^2}\delta(r-vt)rdr + 2\pi\int_{|x|}^{\infty}\frac{\left(1-r^2/v^2t^2\right)^{1/8}}{\left(4\pi vt/(3\Sigma_t)\right)^{3/2}}e^{-v\Sigma_t t}G\left(v\Sigma_t t\left[1-\frac{r^2}{v^2t^2}\right]^{3/4}\right)\Theta(vt-r)rdr \tag{9}$$

$$= \frac{e^{-v\Sigma_t t}}{2vt}\Theta(vt-|x|) + \Theta(vt-|x|)\frac{2\pi e^{-v\Sigma_t t}}{\left(4\pi vt/(3\Sigma_t)\right)^{3/2}}\int_{|x|}^{vt}\left(1-\frac{r^2}{v^2t^2}\right)^{1/8}G\left(v\Sigma_t t\left[1-\frac{r^2}{v^2t^2}\right]^{3/4}\right)rdr \tag{10}$$

$$\overset{\xi=r/vt}{=} \frac{e^{-v\Sigma_t t}}{2vt}\Theta(vt-|x|) + \Theta(vt-|x|)\frac{2\pi e^{-v\Sigma_t t}(vt)^{1/2}}{\left(4\pi/(3\Sigma_t)\right)^{3/2}}\int_{\frac{|x|}{vt}}^{1}\left(1-\xi^2\right)^{1/8}G\left(a\left[1-\xi^2\right]^{3/4}\right)\xi d\xi \tag{11}$$

$$\overset{\substack{u=1-\xi^2\\ w=au^{3/4}}}{=} \frac{e^{-v\Sigma_t t}}{2vt}\left[1+\sqrt{\frac{3}{\pi}}\int_0^{w_0}\sqrt{w}G(w)dw\right]\Theta(vt-|x|) \tag{12}$$

with $w_0 = a\left(1 - \frac{x^2}{(vt)^2}\right)^{3/4}$, the powers combining as $\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$ and the Σ_t dependence cancelling in the last step (I used the assistance of AI in this algebric derivation). Finally, with the interpolation formula of Paasschens, $G(w) = e^w\sqrt{1 + b/w}$ with $b = 2.026$, the integral has a closed form:

$$\int_0^{w_0} e^w \sqrt{w+b}\, dw = e^{w_0}\sqrt{w_0+b} - \sqrt{b} - \frac{\sqrt{\pi}}{2} e^{-b}\left[\operatorname{erfi}\left(\sqrt{w_0+b}\right) - \operatorname{erfi}\left(\sqrt{b}\right)\right] \tag{13}$$

which **closes the derivation of the planar-source scalar flux for $c = 1$.** Of course the relation proved in Q1 can be used to get the solution for $c \neq 1$ using a simple scaling and multiplication by an exponential factor of the solution for $c = 1$.

3 Question 3 - Classical and asymptotic diffusion against the exact solution

3.1 The three solutions being compared

Throughout, $\Sigma_t = v = 1$ as specified in the assignment, the source is the plane pulse $Q(x,t) = \delta(x)\delta(t)$, and the figures are generated by `src/homework2`.

This section covers parts 3(a) and 3(b): all three curves being compared are closed forms, so no numerical scheme enters. Parts 3(c) and 3(d) — the finite-difference time-dependent diffusion code and the relative-error study — are separate deliverables, and the analytic solution derived below is what that code will be verified against.

The source being a pulse in *time* as well as in space is not an assumption but a requirement of the solution being used: Paasschens takes the isotropic point source $S(\mathbf{r}, t, \hat{\mathbf{s}}) = \delta(\mathbf{r})\delta(t)$ (his Eq. 4), and the uncollided term $\propto \delta(r - vt)$ is its signature — every uncollided particle sits at radius exactly vt, which only holds for an instantaneous release. A steady $\delta(\mathbf{r})$ source would give a time-independent $e^{-\Sigma_t r}/(4\pi r^2)$ instead, and there would be no $t = 1, 2, 3, 4, 7, 15$ to present.

The **exact** solution is the planar-source scalar flux derived in Question 2 from the Paasschens point-source solution, carried to general c by the scaling identity of Question 1:

$$\phi(x,t;1) = \frac{e^{-t}}{2t}\left[1 + \sqrt{\frac{3}{\pi}}\int_0^{w_0} \sqrt{w}\, G(w)\, dw\right]\Theta(t - |x|), \qquad w_0 = t\left(1 - \frac{x^2}{t^2}\right)^{3/4} \tag{14}$$

$$\phi(x,t;c) = c\, e^{-(1-c)t}\,\phi(cx, ct; 1) \tag{15}$$

The two **diffusion** solutions are the Green's functions of

$$\frac{1}{v}\frac{\partial\phi}{\partial t} - D\frac{\partial^2\phi}{\partial x^2} + \Sigma_a\phi = \delta(x)\delta(t), \qquad \Sigma_a = (1-c)\Sigma_t, \tag{16}$$

derived in Section 3.2, differing only through $D = D_0(c)/\Sigma_t$:

$$D_0 = \tfrac{1}{3} \quad \text{(classical)}, \qquad D_0(c) = (1-c)\nu_0^2 \quad \text{(asymptotic)}, \tag{17}$$

where ν_0 is the discrete transport eigenvalue, $c\nu_0 \operatorname{arctanh}(1/\nu_0) = 1$. For $c > 1$ the eigenvalue is imaginary, $\nu_0 = i|\nu_0|$, and $(1-c)\nu_0^2 = (c-1)|\nu_0|^2$ remains positive; both branches tend to $\frac{1}{3}$ as $c \to 1$, so the two approximations coincide there.

c	0.6	0.8	1.0	1.2	1.5
$D_0(c)$	0.48588	0.39629	0.33333	0.28717	0.23745

Table 1: The asymptotic diffusion coefficient; the classical value is 1/3 for every c.

3.2 Derivation of the time-dependent diffusion Green's function

3.2.1 The equation, and where the $1/v$ belongs

Written for the *scalar flux* $\phi = vn$, the time-dependent diffusion equation carries a factor $1/v$ on the time derivative:

$$\frac{1}{v}\frac{\partial \phi}{\partial t} - D\frac{\partial^2 \phi}{\partial x^2} + \Sigma_a \phi = S(x,t) \tag{18}$$

The quickest way to see that it must be there is dimensional. With $D = D_0/\Sigma_t$ a length and Σ_a an inverse length, both $D\,\partial_x^2\phi$ and $\Sigma_a\phi$ carry units of ϕ/length; the time derivative carries ϕ/time, and only the factor $1/v$, of units time/length, brings it into line. Equivalently, ∂_x and $\frac{1}{v}\partial_t$ are the pair that appear together in the transport equation from which (18) descends, since a particle covers $v\,dt$ of path in dt of time.

Written instead for the *density* $n = \phi/v$, the same equation reads $\partial_t n - Dv\,\partial_x^2 n + v\Sigma_a n = vS$, with no $1/v$: the factor is a statement about which variable is being used, not an extra physical ingredient. In the dimensionless units of this assignment, $\Sigma_t = v = 1$, the two forms coincide, which is why the distinction never reaches the numbers.

3.2.2 Reduction to the heat equation

Take $S = \delta(x)\delta(t)$ and multiply (18) by v:

$$\frac{\partial \phi}{\partial t} - Dv\frac{\partial^2 \phi}{\partial x^2} + v\Sigma_a \phi = v\,\delta(x)\delta(t) \tag{19}$$

Step 1: the pulse source is an initial condition. Integrate (19) over $t \in (0^-, 0^+)$. The diffusion and absorption terms are bounded and contribute nothing in the limit, so only the time derivative survives:

$$\phi(x,0^+) - \phi(x,0^-) = v\,\delta(x) \qquad \Longrightarrow \qquad \phi(x,0^+) = v\,\delta(x) \tag{20}$$

since nothing precedes the pulse. For $t > 0$ the equation is source-free, so the problem is now a pure initial-value problem. This is the same manoeuvre as in Assignment 1, Question 2, where integrating the steady equation across the source turned the delta into a boundary current instead: a delta source is always better absorbed into a condition than kept as a term.

Step 2: absorption factors out exactly. Absorption removes particles at a rate independent of position, so it can only rescale the solution, never reshape it. Substituting $\phi(x,t) = e^{-v\Sigma_a t}\,u(x,t)$ into (19) for $t > 0$, the $-v\Sigma_a u$ produced by the time derivative cancels the absorption term identically and leaves the pure heat equation

$$\frac{\partial u}{\partial t} = Dv\frac{\partial^2 u}{\partial x^2}, \qquad u(x,0^+) = v\,\delta(x) \tag{21}$$

Note this step is untouched by the sign of $1 - c$: for $c > 1$ the exponential grows instead of decays, and nothing else changes.

Step 3: Fourier transform. With $\hat{u}(k,t) = \int_{-\infty}^{\infty} u(x,t)e^{-ikx}dx$, the spatial derivative becomes multiplication by $-k^2$ and the PDE becomes an ordinary differential equation in t for each k separately:

$$\frac{d\hat{u}}{dt} = -Dvk^2\,\hat{u}, \qquad \hat{u}(k,0^+) = v \qquad \Longrightarrow \qquad \hat{u}(k,t) = v\,e^{-Dvk^2t} \tag{22}$$

the initial value being v because the transform of $v\delta(x)$ is the constant v — a delta excites every wavenumber equally.

Step 4: invert. Completing the square in the exponent, $-Dvtk^2+ikx = -Dvt\left(k - \frac{ix}{2Dvt}\right)^2 - \frac{x^2}{4Dvt}$, and using $\int_{-\infty}^{\infty} e^{-\alpha k^2}dk = \sqrt{\pi/\alpha}$:

$$u(x,t) = \frac{v}{2\pi}\int_{-\infty}^{\infty} e^{-Dvk^2t}e^{ikx}dk = \frac{v}{\sqrt{4\pi Dvt}}\,e^{-x^2/(4Dvt)} \tag{23}$$

Restoring the factor from Step 2 gives the Green's function:

$$\boxed{\phi_{\text{diff}}(x,t;c) = \frac{v\,e^{-(1-c)\Sigma_t vt}}{\sqrt{4\pi Dvt}}\,\exp\left(-\frac{x^2}{4Dvt}\right)} \tag{24}$$

which in the units of this assignment, $\Sigma_t = v = 1$ and $D = D_0(c)$, is the form implemented in `src/homework2/diffusion.py`:

$$\phi_{\text{diff}}(x,t;c) = \frac{e^{-(1-c)t}}{\sqrt{4\pi D_0(c)\,t}}\,e^{-x^2/(4D_0(c)t)} \tag{25}$$

3.2.3 Three checks on the Green's function

Particle balance. Integrating (24) over all x gives $\int \phi\,dx = v\,e^{-(1-c)\Sigma_t vt}$, i.e. $\int n\,dx = e^{-(1-c)t}$ in dimensionless units. This is exactly the normalisation the exact transport solution obeys, by the scaling identity (15) — so the two solutions being compared carry the same number of particles at every time, and the differences in the figures are differences of *shape* alone.

Spreading rate. The Gaussian has variance $\langle x^2\rangle = 2Dvt$, so the particle cloud spreads as $\sqrt{t}$ with no upper bound on speed. The exact solution spreads at most as vt. This is the origin of the leakage past the causal front seen in every panel of Section 3.3, and it is a structural defect of the diffusion approximation, not a small-parameter error.

The steady limit. A steady source is a pulse source fired repeatedly, so integrating (24) over all elapsed times must reproduce the steady solution. Using

$$\int_0^{\infty} \frac{e^{-\beta\tau}}{\sqrt{4\pi a\tau}}e^{-x^2/(4a\tau)}d\tau = \frac{1}{2\sqrt{a\beta}}\,e^{-\sqrt{\beta/a}\,|x|} \tag{26}$$

with $\tau = vt$, $a = D$ and $\beta = \Sigma_a$:

$$\int_0^{\infty} \phi_{\text{diff}}\,dt = \frac{1}{2\sqrt{D\Sigma_a}}e^{-\sqrt{\Sigma_a/D}\,|x|} \overset{D=1/(3\Sigma_t)}{=} \frac{\sqrt{3}}{2\sqrt{1-c}}\,e^{-\sqrt{3(1-c)}\,\Sigma_t|x|} \tag{27}$$

the familiar classical result, and with $D = (1-c)\nu_0^2/\Sigma_t$ instead, $e^{-\Sigma_t|x|/\nu_0}/\big(2(1-c)\nu_0\big)$, the asymptotic one. This ties the coefficient in front of the Gaussian to the coefficient in front of the exponential, so a normalisation slip in either shows up immediately; it is run numerically in `main.py` and agrees to $\sim 10^{-14}$ relative. Note it exists only for $c < 1$: for $c \geq 1$ nothing removes the particles a steady source keeps adding, the integral diverges, and only the time-dependent form (24) is meaningful.

3.3 Parts 3(a) and 3(b): the comparison

Each figure fixes c and shows $t = 1, 2, 3, 4, 7, 15$. The vertical line marks the causal front $|x| = vt$, beyond which the exact solution is identically zero. Only $x \geq 0$ is plotted; all three solutions are even in x.

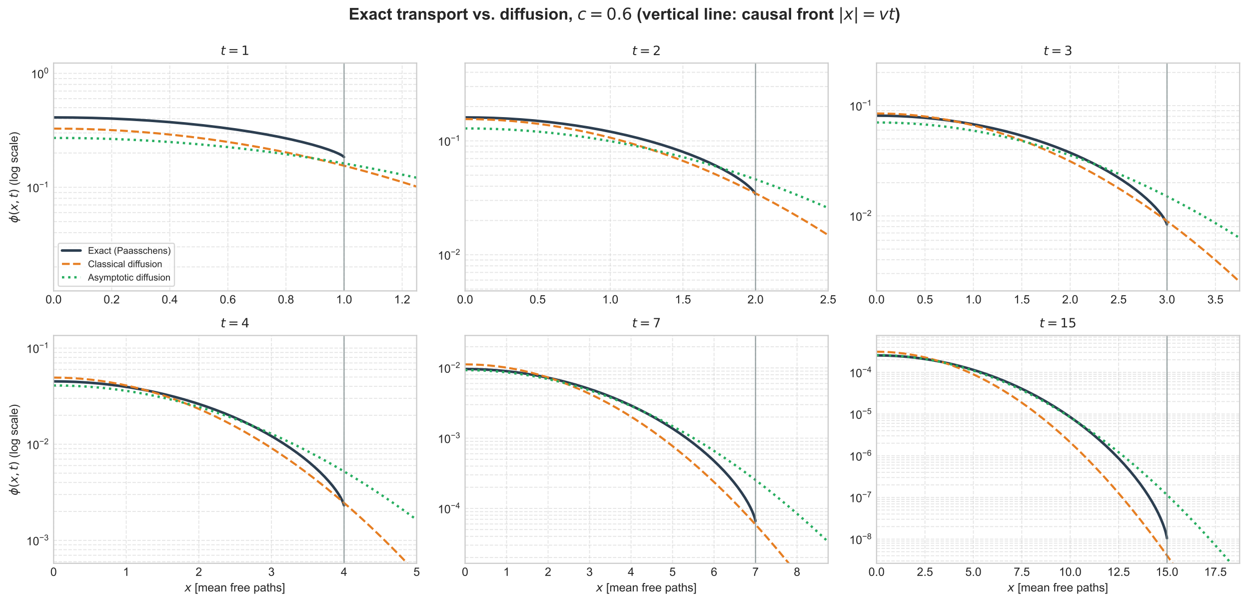

Figure 1: $c = 0.6$.

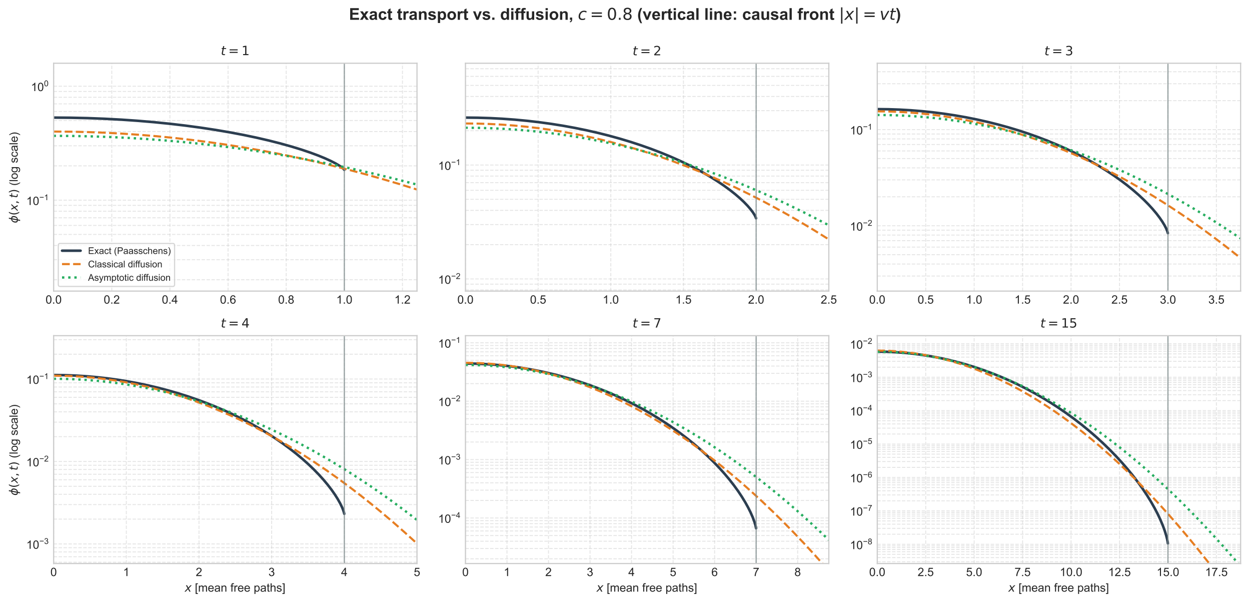

Figure 2: $c = 0.8$.

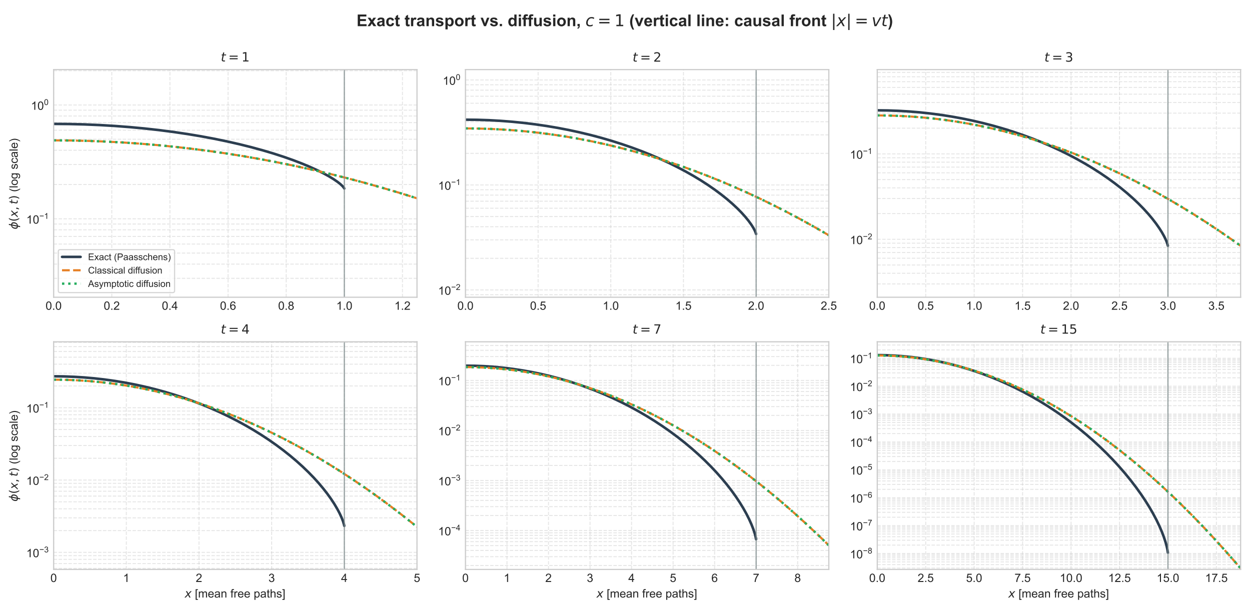

Figure 3: $c = 1$. The two diffusion coefficients coincide at $D_0 = 1/3$, so the classical and asymptotic curves overlie each other.

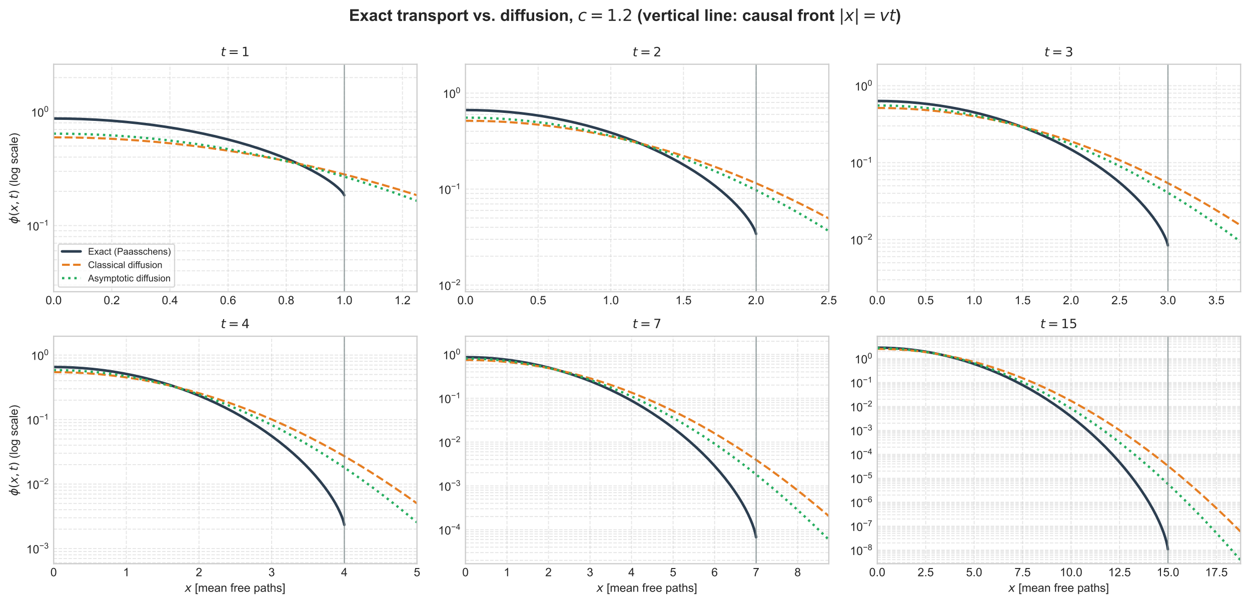

Figure 4: $c = 1.2$.

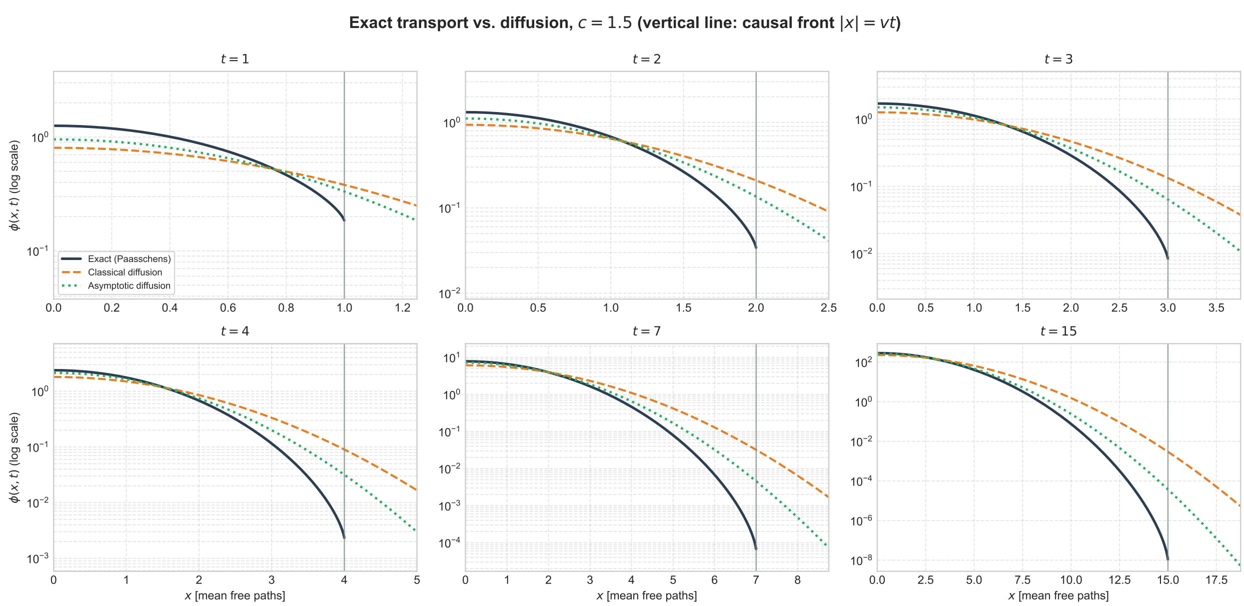

Figure 5: $c = 1.5$.